

Flatness-Based and Passivity-Based Control with Adaptive Neural Compensation of a Direct-Drive PMSG Wind Energy Conversion System

Said Regani
L2EI Laboratory
Jijel University

Manal Messadi
L2EI Laboratory
Jijel University

Kemih Karim
L2EI Laboratory
Jijel University

BP 98 Ouled Aissa, Jijel 18000, Algeria BP 98 Ouled Aissa, Jijel 18000, Algeria BP 98 Ouled Aissa, Jijel 18000, Algeria

Abstract—This paper presents the modeling and control of a grid-connected 2 MW direct-drive permanent magnet synchronous generator (PMSG) wind turbine with a back-to-back converter and an L filter. The model is written in the receptor convention and its signs are verified by an exact power balance. The rotor speed and the energy stored in the DC link and in the filter are used as flat outputs of the two outer loops, and the current loops of both converters are designed by passivity-based control (PBC). A radial basis function (RBF) network, adapted online by a Lyapunov law with σ -modification, learns the uncertainty of the aerodynamic-torque model and of the generator parameters, and its stability is proven. Simulations over 40 s with turbulent wind, parameter mismatch and a 50 % grid voltage dip show that, compared with PI vector control tuned on the same poles, the proposed control reduces the speed tracking error by 24 % and keeps the DC-link voltage within 0.21 V outside the dip (against 9.9 V), while the network reduces the speed error by a further 35 %. The energy captured is the same for all controllers (MPPT efficiency 97.7 %).

Index Terms—PMSG, wind energy conversion, flatness, passivity-based control, RBF neural network, back-to-back converter.

I. INTRODUCTION

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1], [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]. The converter must track the maximum power point, regulate the DC-link voltage and inject the power into the grid at unity power factor.

These tasks are usually performed by cascaded PI loops in rotating frames [5], [6]. The PMSG model is nonlinear and coupled, the aerodynamic torque is not measured, and the parameters vary with temperature and saturation. Nonlinear controllers have therefore been studied for wind turbines and PMSGs, such as backstepping [7], predictive control [8] and passivity-based control [9]. Differential flatness [10] computes the control directly from the desired trajectory of a flat output; with the stored energy as flat output, the DC-link dynamics become a linear power balance [11]. Passivity-based control [12] shapes the energy of the closed loop and keeps the workless coupling terms of the machine. Both methods

use a model: their weak point is the uncertain aerodynamic torque, which neural networks can learn online [13], [14].

This paper applies these tools to the complete chain of a 2 MW direct-drive PMSG turbine:

- a model in the receptor convention whose signs are checked by an exact power balance;
- flat outer loops for the rotor speed and for the energy of the DC link and filter, and PBC current loops for both converters;
- an RBF network that learns the aerodynamic-torque and parameter uncertainty inside the flat speed law, with a Lyapunov stability proof;
- an evaluation under turbulent wind, parameter mismatch and a grid voltage dip against PI vector control tuned on the same poles.

II. WIND TURBINE MODEL

The power captured by the rotor and its tip-speed ratio are [16]

$$P_w = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3, \quad \lambda = \frac{\Omega R}{v} \quad (1)$$

where ρ is the air density, R the rotor radius, v the wind speed, Ω the rotor speed and β the pitch angle ($\beta = 0$ below rated wind). The power coefficient is

$$C_p = 0.5176 \left(\frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-21/\lambda_i} + 0.0068\lambda, \quad (2)$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$; its maximum $C_{p,\max} = 0.48$ is reached at $\lambda_{opt} = 8.1$. The aerodynamic torque is $T_w = P_w/\Omega$ and the direct-drive shaft obeys

$$J \frac{d\Omega}{dt} = T_w + T_e - B\Omega \quad (3)$$

where J is the total inertia, B the friction coefficient and T_e the electromagnetic torque in the receptor convention ($T_e < 0$ in generating mode).

The wind speed combines a mean value, the low-frequency harmonics of [14] and a turbulence:

$$v(t) = V_m + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_t(t) \quad (4)$$

with $V_m = 8.5$ m/s, $A = [1.5, 0.2, 0.3]$ m/s and $f = [0.03, 0.16, 0.17]$ Hz. The rotor-effective turbulence v_t is a first-order coloured noise with standard deviation 0.5 m/s (6 % of V_m) and correlation time 1.5 s. The wind used by the controller is measured by an anemometer modelled by a first-order lag of 1 s, v_m .

III. PMSG, CONVERTERS AND GRID

A. PMSG

In the rotor dq frame (amplitude-invariant transformation, d axis on the magnet flux), with the receptor convention and $L_d = L_q = L_s$ [4],

$$v_{sd} = R_s i_{sd} + L_s \frac{di_{sd}}{dt} - \omega_e L_s i_{sq} \quad (5)$$

$$v_{sq} = R_s i_{sq} + L_s \frac{di_{sq}}{dt} + \omega_e L_s i_{sd} + \omega_e \psi_f \quad (6)$$

$$T_e = \frac{3}{2} p \psi_f i_{sq} \quad (7)$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux. In generating mode $i_{sq} < 0$. With $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd} \ v_{sq}]^T$, $\mathbf{e}_q = [0 \ 1]^T$ and the skew-symmetric matrix $\mathcal{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$,

$$L_s \dot{\mathbf{i}}_s = \mathbf{v}_s - R_s \mathbf{i}_s + \omega_e L_s \mathcal{J} \mathbf{i}_s - \omega_e \psi_f \mathbf{e}_q. \quad (8)$$

B. Back-to-Back Converter, DC Link and Grid

The two-level converters are represented by their average model; the voltage vectors are limited to $V_{dc}/\sqrt{3}$ (space-vector modulation). The DC link receives the power of the machine-side converter (MSC) and supplies the grid-side converter (GSC):

$$CV_{dc} \frac{dV_{dc}}{dt} = P_{dc} - P_i, \quad P_{dc} = -\frac{3}{2} \mathbf{v}_s^T \mathbf{i}_s, \quad P_i = \frac{3}{2} \mathbf{v}_i^T \mathbf{i}_g. \quad (9)$$

In the grid-voltage-oriented frame ($v_{gq} = 0$, ideal PLL), the L filter between the GSC voltage $\mathbf{v}_i$ and the grid voltage $\mathbf{v}_g$, with the current $\mathbf{i}_g$ counted towards the grid, gives

$$L_f \dot{\mathbf{i}}_g = \mathbf{v}_i - R_f \mathbf{i}_g + \omega_g L_f \mathcal{J} \mathbf{i}_g - \mathbf{v}_g, \quad (10)$$

and the grid powers are $P_g = \frac{3}{2} v_{gd} i_{gd}$ and $Q_g = -\frac{3}{2} v_{gd} i_{gq}$.

C. Power Balance

Equations (3), (8)–(10) form a sixth-order model with state $[i_{sd}, i_{sq}, \Omega, V_{dc}, i_{gd}, i_{gq}]$. With the stored energy $E = \frac{1}{2} J \Omega^2 + \frac{3}{4} L_s \|\mathbf{i}_s\|^2 + \frac{1}{2} C V_{dc}^2 + \frac{3}{4} L_f \|\mathbf{i}_g\|^2$, and since $\mathbf{x}^T \mathcal{J} \mathbf{x} = 0$ and $T_e \Omega = \frac{3}{2} \omega_e \psi_f i_{sq}$, the model gives exactly

$$\dot{E} = P_w - B \Omega^2 - \frac{3}{2} R_s \|\mathbf{i}_s\|^2 - \frac{3}{2} R_f \|\mathbf{i}_g\|^2 - P_g \quad (11)$$

for any state and converter voltages: the aerodynamic power is stored, dissipated or delivered to the grid. This identity checks every sign of the model; numerically, its residual is below $10^{-15} P_n$.

Table I lists the parameters. The turbine data (R , Ω_n , T_n) correspond to a 2 MW direct-drive turbine; the generator, DC-link, filter and grid data are those of the non-salient PMSG of [15]. The inertia gives an inertia constant $H =$

TABLE I
PARAMETERS OF THE STUDIED SYSTEM

Parameter	Value	Parameter	Value
P_n	2 MW	p	26
R	40 m	R_s	0.78 m Ω
Ω_n, v_n	2.23 rad/s, 11.0 m/s	L_s	1.57 mH
T_n	0.897 MN·m	ψ_f	9.18 Wb
$\lambda_{opt}, C_{p,max}$	8.1, 0.48	C, V_{dc}^*	20 mF, 1200 V
J	2.8×10^6 kg·m ²	L_f, R_f	0.15 mH, 2 m Ω
B	10^3 N·m·s/rad	Grid	690 V, 50 Hz
ρ	1.225 kg/m ³	f_{sw}	2160 Hz

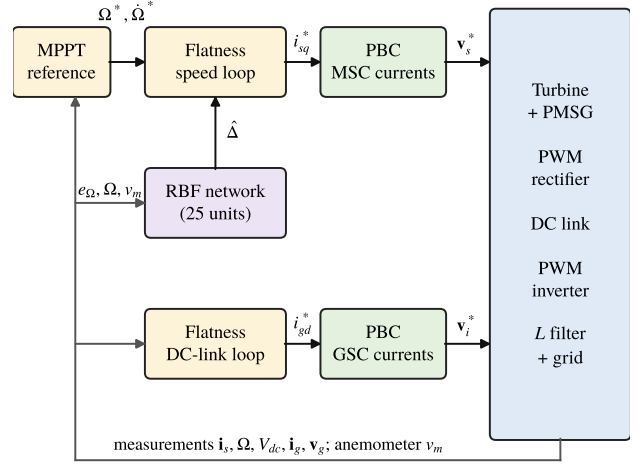


Fig. 1. Control structure: flat outer loops, PBC current loops and RBF compensation.

$J\Omega_n^2/(2P_n) = 3.5$ s. At the rated point ($i_{sq} = -2.51$ kA, $\omega_e = 58$ rad/s), the MSC voltage is 577 V and the GSC voltage 579 V, below the limit $V_{dc}/\sqrt{3} = 693$ V.

IV. PROPOSED CONTROL

The structure is shown in Fig. 1. All errors are defined as measured minus reference values. The controller is sampled at $T_s = 1/(2f_{sw}) = 231.5 \mu\text{s}$ (regular sampling, double update).

A. Speed Reference

The optimal speed $\Omega_{opt} = \lambda_{opt} v_m / R$, limited to $[0.5\Omega_n, \Omega_n]$, is filtered by $\hat{\Omega}^* = \omega_r^2 (\Omega_{opt} - \Omega^*) - 2\omega_r \dot{\Omega}^*$ ($\omega_r = 1$ rad/s), which provides a smooth reference Ω^* and its derivative.

B. Flatness-Based Speed Loop

A system is flat if its states and inputs are functions of a flat output and of a finite number of its derivatives [10]. Flatness is used here for the two outer subsystems, with the currents as inputs. From (3), $T_e = J\dot{\Omega} + B\Omega - T_w$: Ω is a flat output of the shaft with input T_e , i.e., i_{sq} . Imposing $\hat{\Omega} = \nu_1$ with $\nu_1 = \hat{\Omega}^* - k_{11}e_\Omega - k_{12}\int e_\Omega dt$, $e_\Omega = \Omega - \Omega^*$, gives

$$T_e^* = J\nu_1 + B\Omega - \hat{T}_w - \hat{\Delta}, \quad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f}, \quad i_{sd}^* = 0 \quad (12)$$

where $\hat{T}_w$ is the aerodynamic torque computed from (1)–(2) with v_m and $\hat{\Delta}$ is the neural compensation (Section IV-E). T_e^* is limited to $[-1.2T_n, 0]$ and the integral is frozen at the limit. With $i_{sq} = i_{sq}^*$ and $\Delta = T_w - \hat{T}_w$,

$$J(\dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega dt) = \Delta - \hat{\Delta}. \quad (13)$$

C. Flatness-Based DC-Link Loop

The flat output is the energy stored in the DC link and in the filter, $y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$. From (9)–(10),

$$\dot{y}_2 = P_{dc} - P_g - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 \quad (14)$$

which is linear in the grid power [11]; V_{dc} follows from y_2 and the measured currents, and P_g from $\dot{y}_2$. Imposing $\dot{y}_2 = -k_{21}e_y - k_{22}\int e_y dt$, with $e_y = y_2 - y_2^*$ and $y_2^* = \frac{1}{2}CV_{dc}^{*2} + \frac{3}{4}L_f\|\mathbf{i}_g^*\|^2$, gives

$$P_g^* = P_{dc} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 + k_{21}e_y + k_{22}\int e_y dt, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}} \quad (15)$$

and $i_{gq}^* = -2Q_g^*/(3v_{gd})$ with $Q_g^* = 0$. P_{dc} is computed from the applied MSC voltages and the measured currents, so the power coming from the generator is compensated before it changes V_{dc} . The variation of y_2^* , due to the filter energy only (below 300 J against 14.4 kJ in the capacitor), is neglected. The gains are $k_{11} = 2\omega_\Omega$, $k_{12} = \omega_\Omega^2$, $k_{21} = 2\omega_v$, $k_{22} = \omega_v^2$ (Table II).

D. Passivity-Based Current Loops

With $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$ and $\dot{\mathbf{z}}_s = \mathbf{e}_s$, the MSC voltage reference is

$$\mathbf{v}_s^* = L_s\dot{\mathbf{i}}_s^* + R_s\mathbf{i}_s^* - \omega_e L_s \mathcal{J}\mathbf{i}_s^* + \omega_e \psi_f \mathbf{e}_q - R_a \mathbf{e}_s - K_i \mathbf{z}_s \quad (16)$$

and, with $\mathbf{e}_g = \mathbf{i}_g - \mathbf{i}_g^*$ and $\dot{\mathbf{z}}_g = \mathbf{e}_g$, the GSC voltage reference is

$$\mathbf{v}_i^* = L_f\dot{\mathbf{i}}_g^* + R_f\mathbf{i}_g^* - \omega_g L_f \mathcal{J}\mathbf{i}_g^* + \mathbf{v}_g - R_b \mathbf{e}_g - K_{ig} \mathbf{z}_g. \quad (17)$$

The reference derivatives are obtained by first-order filters (2 ms); the integrals are frozen when the voltage reaches $V_{dc}/\sqrt{3}$.

Proposition 1. *With (16) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2}L_s\|\mathbf{e}_s\|^2 + \frac{1}{2}K_i\|\mathbf{z}_s\|^2$ satisfies $\dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2$ and $\mathbf{e}_s \rightarrow 0$. The same holds for (17) applied to (10).*

Proof. Subtracting $L_s\dot{\mathbf{i}}_s^*$ from (8) and inserting (16) gives $L_s\dot{\mathbf{e}}_s = -(R_s + R_a)\mathbf{e}_s + \omega_e L_s \mathcal{J}\mathbf{e}_s - K_i \mathbf{z}_s$. Hence $\dot{H} = \mathbf{e}_s^T L_s \dot{\mathbf{e}}_s + K_i \mathbf{z}_s^T \dot{\mathbf{e}}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2$, since $\mathbf{e}_s^T \mathcal{J}\mathbf{e}_s = 0$: the coupling is workless. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [17] gives $\mathbf{e}_s \rightarrow 0$. $\square$

The gains place a double closed-loop pole at ω_i : $R_s + R_a = 2\omega_i L_s$, $K_i = \omega_i^2 L_s$, $R_f + R_b = 2\omega_i L_f$, $K_{ig} = \omega_i^2 L_f$. With $\omega_i = 500$ rad/s, the current loops are 250 times faster than the speed loop and 8 times faster than the DC-link loop.

TABLE II
CONTROLLER PARAMETERS

Loop	Parameters
Speed (flat)	$\omega_\Omega = 2$ rad/s ($k_{11} = 4$, $k_{12} = 4$), $\omega_r = 1$ rad/s
DC link (flat)	$\omega_v = 60$ rad/s ($k_{21} = 120$, $k_{22} = 3600$)
PBC MSC	$\omega_i = 500$ rad/s: $R_a = 1.57 \Omega$, $K_i = 393 \Omega/\text{s}$
PBC GSC	$\omega_i = 500$ rad/s: $R_b = 0.148 \Omega$, $K_{ig} = 37.5 \Omega/\text{s}$
RBF network	25 units, $b = 0.15$, $\Gamma = 10^7$, $\sigma = 0.02 \text{ s}^{-1}$

E. Adaptive RBF Compensation

The torque model error $\Delta = T_w - \hat{T}_w$ comes from the C_p model, the anemometer and the generator parameters. On the operating region it is approximated by $\Delta = \mathbf{W}^{*T}\phi(\mathbf{z}) + \varepsilon$, $|\varepsilon| \leq \bar{\varepsilon}$, where $\mathbf{z} = [\Omega/\Omega_n, v_m/v_n]^T$ and ϕ contains 25 Gaussian functions $\phi_j = \exp(-\|\mathbf{z} - \mathbf{c}_j\|^2/(2b^2))$ with centres on a 5×5 grid of $[0.5, 1.1]^2$ and $b = 0.15$. The compensation $\hat{\Delta} = \hat{\mathbf{W}}^T\phi$ replaces the integral of the speed loop ($k_{12} = 0$ when the network is used) and is adapted by

$$\dot{\hat{\mathbf{W}}} = \Gamma(\phi e_\Omega - \sigma \hat{\mathbf{W}}), \quad (18)$$

frozen when T_e^* saturates, with $\hat{\mathbf{W}}(0) = 0$.

Proposition 2. *Consider (13) with $k_{12} = 0$, the law (18) with $\Gamma, \sigma > 0$, exact J and B , and no saturation. Then e_Ω and $\tilde{\mathbf{W}} = \mathbf{W}^* - \hat{\mathbf{W}}$ are uniformly ultimately bounded; the residual set shrinks when k_{11} increases and when $\bar{\varepsilon}$ and $\sigma\|\mathbf{W}^*\|^2$ decrease.*

Proof. With $V = \frac{1}{2}Je_\Omega^2 + \frac{1}{2\Gamma}\|\tilde{\mathbf{W}}\|^2$ and (13), $\dot{V} = -Jk_{11}e_\Omega^2 + e_\Omega(\tilde{\mathbf{W}}^T\phi + \varepsilon) - \frac{1}{\Gamma}\tilde{\mathbf{W}}^T\dot{\tilde{\mathbf{W}}} = -Jk_{11}e_\Omega^2 + e_\Omega\varepsilon + \sigma\tilde{\mathbf{W}}^T\tilde{\mathbf{W}}$. Using $\tilde{\mathbf{W}}^T\tilde{\mathbf{W}} \leq \frac{1}{2}\|\mathbf{W}^*\|^2 - \frac{1}{2}\|\tilde{\mathbf{W}}\|^2$ and $e_\Omega\varepsilon \leq \frac{Jk_{11}}{2}e_\Omega^2 + \frac{\bar{\varepsilon}^2}{2Jk_{11}}$ gives $\dot{V} \leq -\frac{Jk_{11}}{2}e_\Omega^2 - \frac{\sigma}{2}\|\tilde{\mathbf{W}}\|^2 + \frac{\bar{\varepsilon}^2}{2Jk_{11}} + \frac{\sigma}{2}\|\mathbf{W}^*\|^2$, which is negative outside a compact set. $\square$

The result is local: it assumes fast current loops, no saturation and $\mathbf{z}$ in the region covered by the centres; mismatches of J and B add bounded terms to ε . The PBC loops and the DC-link loop are not modified by the network.

V. SIMULATION RESULTS

A. Setup

The model is simulated in MATLAB/GNU Octave with a fourth-order Runge–Kutta scheme at the sampling period T_s , the converter voltages being applied one sample after their computation. Each run lasts 40 s and starts from the MPPT equilibrium at $v(0)$. Two cases are considered:

- *nominal*: the controller uses the true parameters;
- *robustness*: the true plant has $C_p \times 1.1$, $J \times 1.2$, $R_s \times 1.5$, $L_s \times 1.2$, $\psi_f \times 0.95$, $R_f \times 1.5$, $L_f \times 1.25$, and the grid voltage drops by 50 % between 25 and 25.15 s.

The proposed control, without (flat+PBC) and with the network (flat+PBC+ANN), is compared with PI vector control [5], [6]: PI speed loop, PI loop on the DC-link energy, and PI current loops with dq decoupling and back-EMF/grid-voltage feedforward, tuned on the same poles (ω_Ω , ω_v , ω_i)

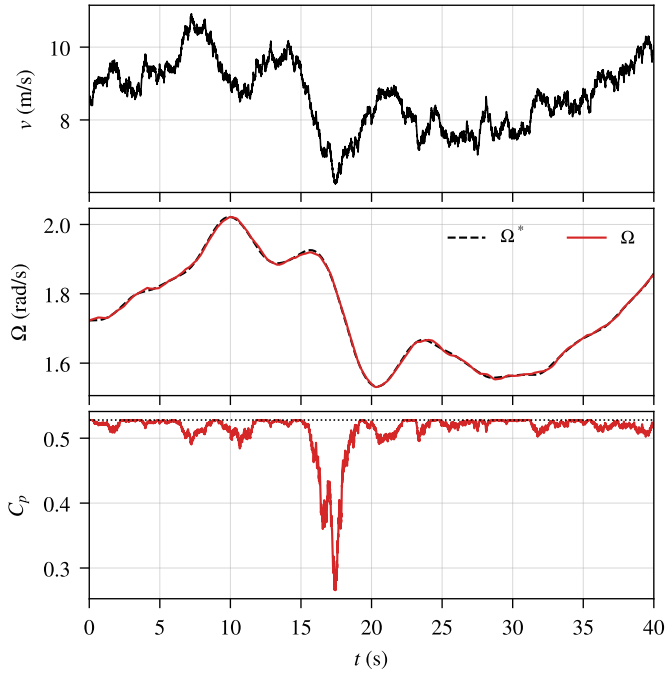


Fig. 2. Robustness case, flat+PBC+ANN: wind speed, rotor speed and reference, and power coefficient (dotted: $C_{p,max}$ of the true turbine).

TABLE III
COMPARISON OF THE CONTROLLERS (40 s, FROM $t = 1$ s)

Metric	PI	flat+PBC	flat+PBC+ANN
<i>Nominal</i>			
IAE of Ω (rad)	0.195	0.148	0.099
RMS error of i_{sq} (A)	0.057	0.044	0.042
Max $ V_{dc} - V_{dc}^* $ (V)	9.7	0.13	0.13
Energy to grid (kWh)	10.255	10.259	10.253
MPPT efficiency (%)	97.7	97.7	97.7
Torque rate, std (MN·m/s)	0.166	0.178	0.194
<i>Robustness</i>			
IAE of Ω (rad)	0.237	0.179	0.117
RMS error of i_{sq} (A)	0.063	0.048	0.045
Max $ V_{dc} - V_{dc}^* $, no dip (V)	9.9	0.21	0.21
Max $ V_{dc} - V_{dc}^* $, dip (V)	24.9	26.0	27.2
Energy to grid (kWh)	11.206	11.211	11.208
MPPT efficiency (%)	97.7	97.7	97.7
Torque rate, std (MN·m/s)	0.170	0.180	0.199

and with the same limits. The PI control has no aerodynamic-torque or P_{dc} feedforward.

B. Results

Figs. 2–5 and Table III summarize the results. The wind varies between 6.2 and 10.9 m/s (Fig. 2). The rotor speed follows its reference; C_p stays close to its maximum except during the fast wind decrease around 17 s, where the large inertia prevents the rotor from slowing down quickly. The electromagnetic torque is negative and follows the aerodynamic torque, i_{sd} is kept at zero, and the grid power follows the captured power with $Q_g = 0$ (Fig. 3).

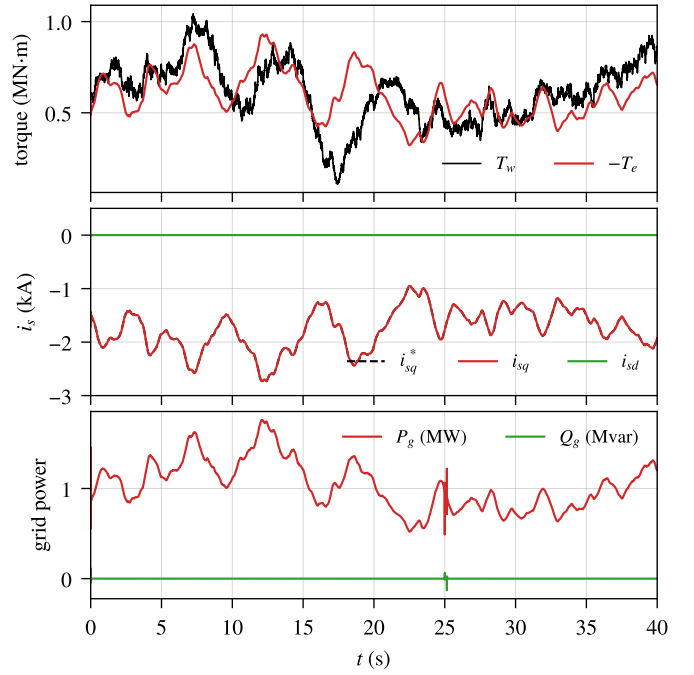


Fig. 3. Robustness case, flat+PBC+ANN: aerodynamic and electromagnetic torques, stator currents, and grid active and reactive powers.

Speed loop. The feedforward of $J\dot{\Omega}^*$ and $\hat{T}_w$ in the flat law reduces the IAE of the speed by 24 % (nominal) and 24 % (robustness) with respect to PI. The network reduces it by a further 34 % and 35 %: it learns the mean uncertainty Δ , about 87 kN·m in the robustness case (15 % of the mean aerodynamic torque), within a few seconds, and follows its slow variations (Fig. 5). The price is a more active torque (11 % larger standard deviation of its rate). The captured energy and the MPPT efficiency are almost identical for the three controllers: near the optimum, C_p is flat, so small speed errors have little effect on the energy.

DC link. With the feedforward of P_{dc} , the flat law keeps V_{dc} within 0.21 V of 1200 V outside the dip, against 9.9 V with the PI loop, which reacts only to the voltage error. During the 50 % dip, the three controllers give a similar deviation of about 26 V (2 % of V_{dc}^*), recovered within 50 ms (Fig. 4): it is set by the current-loop dynamics, the GSC current doubling to keep the same power (Fig. 5).

Currents. The PBC and PI current loops have the same poles and both track the references with errors below 0.1 A RMS for i_{sq} , i.e., less than 0.01 % of the rated current.

These results are obtained with the average converter model and an ideal PLL; the switching ripple, the dead times and the PLL dynamics are not represented.

VI. CONCLUSION

A 2 MW direct-drive PMSG wind energy conversion system was modeled in the receptor convention, with an exact power balance that verifies the signs of the model. Its control combines flatness for the speed and DC-link energy loops,

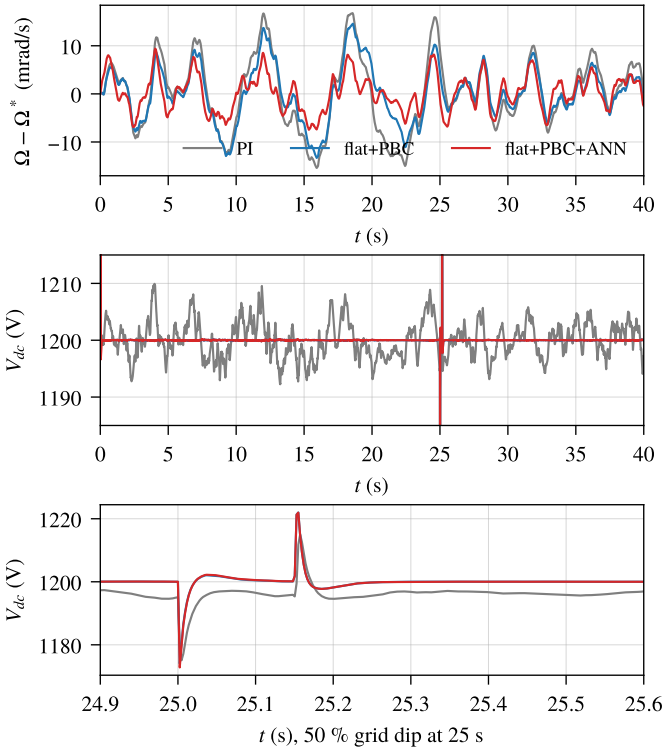


Fig. 4. Robustness case: speed tracking error and DC-link voltage of the three controllers, and DC-link voltage during the 50% grid voltage dip.

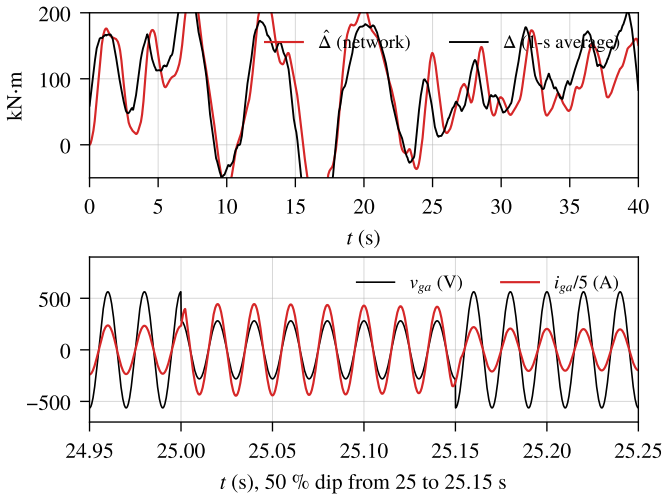


Fig. 5. Robustness case: network output $\hat{\Delta}$ and actual uncertainty Δ (aerodynamic model and flux mismatch, 1-s average), and grid voltage and current of phase a during the dip.

passivity-based control for the current loops of both converters, and an RBF network that learns the aerodynamic-torque and parameter uncertainty with a Lyapunov law. Under turbulent wind and parameter mismatch, the flat structure reduces the speed tracking error with respect to PI vector control and keeps the DC-link voltage almost constant thanks to its power feedforward; the network further reduces the speed error at

the price of a more active torque. The captured energy is the same for all controllers. Future work will include a switched converter model, the PLL, pitch control above rated wind and experimental validation.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

REFERENCES

- [1] International Energy Agency (IEA), *World Energy Outlook 2024*, Paris, France, 2024.
- [2] Global Wind Energy Council (GWEC), *Global Wind Report 2024*, Brussels, Belgium, 2024.
- [3] H. Polinder, J. A. Ferreira, B. B. Jensen, A. B. Abrahamsen, K. Atallah, and R. A. McMahon, "Trends in wind turbine generator systems," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 1, no. 3, pp. 174–185, Sep. 2013.
- [4] B. Wu, Y. Lang, N. Zargari, and S. Kouro, *Power Conversion and Control of Wind Energy Systems*. Hoboken, NJ, USA: Wiley-IEEE Press, 2011.
- [5] M. Chinchilla, S. Arnaltes, and J. C. Burgos, "Control of permanent-magnet generators applied to variable-speed wind-energy systems connected to the grid," *IEEE Trans. Energy Convers.*, vol. 21, no. 1, pp. 130–135, Mar. 2006.
- [6] A. Yazdani and R. Iravani, *Voltage-Sourced Converters in Power Systems: Modeling, Control, and Applications*. Hoboken, NJ, USA: Wiley-IEEE Press, 2010.
- [7] A. Mechter, K. Kemih, and M. Ghanes, "Backstepping control of a wind turbine for low wind speeds," *Nonlinear Dynamics*, vol. 84, no. 4, pp. 2435–2445, 2016.
- [8] M. Messadi, A. Mellit, K. Kemih, and M. Ghanes, "Predictive control of a chaotic permanent magnet synchronous generator in a wind turbine system," *Chinese Phys. B*, vol. 24, no. 1, Art. no. 010502, 2015, doi: 10.1088/1674-1056/24/1/010502.
- [9] H. Takhi, K. Kemih, L. Moysis, *et al.*, "Passivity based control and synchronization of perturbed uncertain chaotic systems and their micro-controller implementation," *Int. J. Dyn. Control*, vol. 8, no. 3, pp. 973–990, 2020.
- [10] M. Fliess, J. Lévine, P. Martin, and P. Rouchon, "Flatness and defect of non-linear systems: Introductory theory and examples," *Int. J. Control*, vol. 61, no. 6, pp. 1327–1361, 1995.
- [11] P. Thounthong, S. Pierfederici, and B. Davat, "Analysis of differential flatness-based control for a fuel cell hybrid power source," *IEEE Trans. Energy Convers.*, vol. 25, no. 3, pp. 909–920, Sep. 2010.
- [12] R. Ortega, A. Loría, P. J. Nicklasson, and H. Sira-Ramírez, *Passivity-Based Control of Euler-Lagrange Systems*. London, UK: Springer, 1998.
- [13] F. L. Lewis, S. Jagannathan, and A. Yeşildirek, *Neural Network Control of Robot Manipulators and Nonlinear Systems*. London, UK: Taylor & Francis, 1999.
- [14] S. Regani, M. Messadi, and K. Kemih, "A novel chaos control approach with adaptive ANN compensation for a DFIG-based wind turbine," *Mathematics*, under review, 2026.
- [15] S. M. M. Hasan and A. H. M. Shatil, "Design and comparison of grid connected permanent magnet synchronous generator non-salient pole and salient pole rotor wind turbine," *AIUB J. Sci. Eng.*, vol. 20, no. 2, pp. 40–46, 2021.
- [16] S. Heier, *Grid Integration of Wind Energy: Onshore and Offshore Conversion Systems*, 3rd ed. Chichester, UK: Wiley, 2014.
- [17] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002.